



PayZo is a Tunisian digital-banking platform that unifies multi-bank account management and protects peer-to-peer transfers through a three-layer machine-learning fraud detection system. The primary scorer is a calibrated LightGBM model (PR-AUC 0.90, ROC-AUC 0.99); a Random Forest backup and a rule-based firewall ensure the platform always produces a risk decision. High-risk transfers are suspended and routed to an analyst review queue—a human makes the final call on every fraud alert.

Keywords: digital banking · fraud detection · machine learning · P2P transfer · Tunisia



Tunisian Republic
Ministry of Higher Education and Scientific Research
University of Monastir
Faculty of Sciences of Monastir



FINAL YEAR INTERNSHIP REPORT
in fulfilment of the requirements for the Bachelor's Degree
in Software Engineering and Information Systems



Fraud Detection

in Banking Transactions

PREPARED BY
Mohamed Ben Salem

Defended on 13 June 2026 at 12:30, Amphi 3

JURY
President **Mohamed Neji Maatouk**
Reviewer **Nizar Kerkeni**
Academic Supervisor **Mohsen Maraoui**
Professional Supervisor **Chaima Zaier**



ACADEMIC YEAR
2025 / 2026

